3D Vectors: The Vector Product

Questions 1–49

Question 1. Find:

a) $\mathbf{i} \times \mathbf{i}$
b) $\mathbf{i} \times \mathbf{j}$
c) $\mathbf{i} \times \mathbf{k}$
d) $\mathbf{j} \times \mathbf{i}$
e) $\mathbf{j} \times \mathbf{j}$
f) $\mathbf{j} \times \mathbf{k}$
g) $\mathbf{k} \times \mathbf{i}$
h) $\mathbf{k} \times \mathbf{j}$
i) $\mathbf{k} \times \mathbf{k}$

Question 2. Use the distributive property of vector products to find:

a) $\mathbf{i} \times (\mathbf{j} + \mathbf{k})$
b) $\mathbf{j} \times (\mathbf{i} + \mathbf{k})$
c) $\mathbf{k} \times (\mathbf{i} + \mathbf{j})$
d) $(\mathbf{i} - \mathbf{j}) \times (\mathbf{j} - \mathbf{k})$
e) $(\mathbf{k} - \mathbf{i}) \times (\mathbf{j} - \mathbf{i})$
f) $(\mathbf{i} + \mathbf{j} + \mathbf{k}) \times (-\mathbf{i} - \mathbf{j} + \mathbf{k})$
g) $(\mathbf{k} - \mathbf{j} - \mathbf{i}) \times (\mathbf{j} - \mathbf{i} - \mathbf{k})$
h) $(2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}) \times (3\mathbf{i} + \mathbf{j} - \mathbf{k})$
i) $(3\mathbf{j} - 5\mathbf{i} - 3\mathbf{k}) \times (6\mathbf{i} - 8\mathbf{k})$

Question 3. It is given that $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \\ 1 \end{pmatrix}$.

a) Show that $\begin{pmatrix} 8 \\ -5 \\ 6 \end{pmatrix}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

b) Find $\mathbf{a} \times \mathbf{b}$.

Question 4. It is given that $\mathbf{a} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$.

a) Find $\mathbf{a} \times \mathbf{b}$.

b) Explain the geometrical relationship between $\mathbf{a}$ and $\mathbf{b}$ and the answer to part (a).

Question 5. Find:

a) $\begin{pmatrix}1\\2\\1\end{pmatrix} \times \begin{pmatrix}3\\1\\2\end{pmatrix}$

b) $\begin{pmatrix}5\\3\\2\end{pmatrix} \times \begin{pmatrix}4\\1\\6\end{pmatrix}$

c) $\begin{pmatrix}-2\\3\\-9\end{pmatrix} \times \begin{pmatrix}-10\\0\\7\end{pmatrix}$

d) $(3\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}) \times (5\mathbf{i} - \mathbf{j} + \mathbf{k})$

e) $(-10\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) \times (-\mathbf{i} + 5\mathbf{j} + 4\mathbf{k})$

f) $(3\mathbf{i} + 7\mathbf{k}) \times (-2\mathbf{j} - \mathbf{k})$

g) $5\mathbf{i} \times (2\mathbf{i} - 3\mathbf{j} + \mathbf{k})$

h) $(10\mathbf{k} - 3\mathbf{i}) \times 3\mathbf{j}$

Question 6. Find a unit vector that is perpendicular to both $\mathbf{a} = \begin{pmatrix}2\\0\\-3\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}0\\-3\\-1\end{pmatrix}$.

Question 7. Find a unit vector that is perpendicular to both $\mathbf{a} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix}$.

Question 8. Find, in terms of the constant a, a vector that is perpendicular to both $\begin{pmatrix} 1 \\ a \\ 2 \end{pmatrix}$ and $\begin{pmatrix} a \\ 2 \\ 3 \end{pmatrix}$.

Question 9. Find, in terms of the constant p, a vector that is perpendicular to both $\begin{pmatrix} p \\ 3 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 1-p \\ 2 \\ p \end{pmatrix}$.

Question 10. Find, in terms of the constant q, a vector that is perpendicular to both $\begin{pmatrix} 1 \\ q \\ 1-q \end{pmatrix}$ and $\begin{pmatrix} q-1 \\ 1 \\ q \end{pmatrix}$.

Question 11. It is given that $\mathbf{a} = \begin{pmatrix} 2 \\ -7 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -8 \\ -11 \end{pmatrix}$.

a) Find $\mathbf{a} \times \mathbf{b}$.
b) Find $\mathbf{b} \times \mathbf{a}$.
c) Hence write a statement linking your answers to parts (a) and (b).

Question 12. Use the vector product to show that the vectors $\begin{pmatrix} 3 \\ -6 \\ 12 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 8 \\ -16 \end{pmatrix}$ are parallel.

Question 13. OA and OB are two sides of a triangle. It is given that $\overrightarrow{OA} \times \overrightarrow{OB} = \sqrt{2}\mathbf{i} - 9\mathbf{j} - \sqrt{5}\mathbf{k}$. Find the area of the triangle OAB.

Question 14. The position vectors $\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 8 \\ 3 \\ 5 \end{pmatrix}$ form two sides of a parallelogram. Find the area of the parallelogram.

Question 15. It is given that A and B have coordinates $(3, 6, 12)$ and $(5, -10, 20)$ respectively. OA and OB, where O is the origin, form two sides of the parallelogram with vertices O, A, B and C.

a) Find the coordinates of C.

b) Find the area of the parallelogram.

Question 16. It is given that A and B have coordinates $(3, -6, -2)$ and $(4, -6, 0)$ respectively. Find the exact area of the triangle OAB, where O is the origin.

Question 17. Find the exact area of the triangle with vertices:

a) $(-1, 0, 1)$, $(2, 1, 0)$ and $(1, 1, -1)$

b) $(3, -8, 1)$, $(2, 1, -3)$ and $(1, 2, 3)$

c) $(4, 2, -6)$, $(3, 0, -1)$ and $(8, 3, -5)$

d) $(-1, 4, 5)$, $(9, -2, -3)$ and $(0, 5, 5)$

Question 18. The points A, B and C have coordinates $(5, 2, -1)$, $(9, -4, 2)$ and $(-17, 16, -8)$ respectively.

a) Show that the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are linearly dependent, where O is the origin.

b) Find the area of the triangle ABC.

Question 19. The points A, B and C have coordinates $(2, -5, 6)$, $(4, -3, -1)$ and $(-22, 6, 24)$. Show that the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are linearly independent, where O is the origin.

Question 20. Determine whether the following vectors are coplanar:

a) $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$

b) $\begin{pmatrix} -4 \\ 3 \\ 6 \end{pmatrix}$, $\begin{pmatrix} -5 \\ 3 \\ -8 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -9 \\ -144 \end{pmatrix}$

Question 21. It is given that $\mathbf{a} = \begin{pmatrix} -4 \\ 8 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -6 \\ 3 \end{pmatrix}$.

a) Find $\mathbf{a} \times \mathbf{b}$.

b) Hence find the exact value of $\sin\theta$, where θ is the acute angle between vectors $\mathbf{a}$ and $\mathbf{b}$.

Question 22. Use the vector product to find the acute angle between the following pairs of vectors:

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$

b) $\mathbf{a} = \begin{pmatrix} -7 \\ 12 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -20 \\ -1 \end{pmatrix}$

c) $\mathbf{a} = \begin{pmatrix} 2\sqrt{2} \\ 0 \\ 7\sqrt{3} \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5\sqrt{2} \\ 2\sqrt{2} \\ 0 \end{pmatrix}$

d) $\mathbf{a} = \mathbf{i} + 9\mathbf{k}$ and $\mathbf{b} = -\mathbf{j} - 15\mathbf{k}$

e) $\mathbf{a} = \sqrt{2}\mathbf{i} + \sqrt{3}\mathbf{j}$ and $\mathbf{b} = \sqrt{3}\mathbf{i} + \sqrt{2}\mathbf{j} + \sqrt{6}\mathbf{k}$

f) $\mathbf{a} = \pi\mathbf{i} - \sqrt{2}\mathbf{j}$ and $\mathbf{b} = \sqrt{5}\mathbf{i} + \mathbf{j} + \pi\mathbf{k}$

Question 23. Find the value of the constant a given that $\begin{pmatrix} a \\ a \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} -22 \\ 7 \\ 9 \end{pmatrix}$.

Question 24. Find the values of the constants p and q given that $\begin{pmatrix} p \\ 3 \\ -2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 16 \\ q \\ -2 \end{pmatrix}$.

Question 25. Find the values of the constants m and n given that $\begin{pmatrix} 8 \\ 2m \\ -9 \end{pmatrix} \times \begin{pmatrix} 5 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} -45 \\ -37 \\ 3n \end{pmatrix}$.

Question 26. Find the values of the constants p, q and r given that $\begin{pmatrix} p \\ -3 \\ 8 \end{pmatrix} \times \begin{pmatrix} 10 \\ 5 \\ 2q \end{pmatrix} = \begin{pmatrix} -16 \\ 168 \\ 5r \end{pmatrix}$.

Question 27. It is given that $\mathbf{a} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$, $\mathbf{b} = (q-1)\mathbf{i} + q\mathbf{j} - 3\mathbf{k}$ and $\mathbf{c} = 2\mathbf{i} + (2q-3)\mathbf{j} + q\mathbf{k}$, where q is a constant. Given also that $\mathbf{a}$ is perpendicular to $\mathbf{b} \times \mathbf{c}$, find the possible values of q.

Question 28. It is given that for vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, it is such that $\mathbf{a} \times \mathbf{b} = \mathbf{c}$. Express each of these in terms of $\mathbf{c}$:

a) $(2\mathbf{a}) \times (5\mathbf{b})$

b) $(\frac{1}{2}\mathbf{b}) \times (8\mathbf{a})$

Question 29. It is given that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are vectors such that $\mathbf{a}$ is parallel to both $\mathbf{b}$ and $\mathbf{c}$. Find $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$.

Question 30. It is given that $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors. Simplify each of these:

a) $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b})$

b) $(2\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + 3\mathbf{b})$

c) $(5\mathbf{b} - 6\mathbf{a}) \times (6\mathbf{b} - 5\mathbf{a})$

Question 31. It is given that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-parallel vectors. Simplify each of these:

a) $(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{c})$

b) $(\mathbf{c}-\mathbf{b})\times(\mathbf{b}-\mathbf{a})$

c) $(\mathbf{a}-\mathbf{b}+\mathbf{c})\times(\mathbf{b}-\mathbf{a}+\mathbf{c})$

d) $(2\mathbf{a}+\mathbf{b}-3\mathbf{c})\times(4\mathbf{a}-3\mathbf{b}+\mathbf{c})$

Question 32. It is given that $\mathbf{a}$ and $\mathbf{b}$ are vectors such that $(2\mathbf{a}-\mathbf{b})\times(6\mathbf{a}-5\mathbf{b})=\mathbf{0}$. Explain what you can deduce about the vectors $\mathbf{a}$ and $\mathbf{b}$.

Question 33. It is given that the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have the properties $\mathbf{a} \times \mathbf{b} = \mathbf{k}$, $\mathbf{a} \times \mathbf{c} = \mathbf{j}$ and $\mathbf{b} \times \mathbf{c} = \mathbf{i}$. Simplify each of these:

a) $(\mathbf{a} + 2\mathbf{b}) \times (\mathbf{a} - \mathbf{c})$

b) $(2\mathbf{a} + \mathbf{b} + \mathbf{c}) \times (\mathbf{b} + \mathbf{c})$

c) $(3\mathbf{a} - 2\mathbf{b} - \mathbf{c}) \times (\mathbf{c} - 4\mathbf{b} - 3\mathbf{a})$

Question 34. It is given that the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have the properties $\mathbf{a} \times \mathbf{b} = 3\mathbf{i}$ and $\mathbf{a} \times \mathbf{c} = 4\mathbf{j}$. Simplify $(2\mathbf{a} - 3\mathbf{b} + 8\mathbf{c}) \times (7\mathbf{a} + 6\mathbf{b} - 16\mathbf{c})$.

Question 35. It is given that the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have the properties $\mathbf{a} \times \mathbf{b} = 2\mathbf{k}$, $\mathbf{a} \times \mathbf{c} = 5\mathbf{i}$ and $\mathbf{b} \times \mathbf{c} = 3\mathbf{j}$. Simplify $(\mathbf{c} - \mathbf{b} - 4\mathbf{a}) \times (4\mathbf{a} - 6\mathbf{b} + 9\mathbf{c})$.

Question 36. It is given that the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have the properties $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 4 \\ 1 \\ -3 \end{pmatrix}$ and $\mathbf{a} \times \mathbf{c} = \begin{pmatrix} 8 \\ -3 \\ 2 \end{pmatrix}$. Simplify $(-2\mathbf{a} - 6\mathbf{b} + 3\mathbf{c}) \times 5\mathbf{a}$.

Question 37. It is given that the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have the properties $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$, $\mathbf{a} \times \mathbf{c} = \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix}$ and $\mathbf{b} \times \mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$. Simplify $(3\mathbf{a} - \mathbf{b} + 2\mathbf{c}) \times (3\mathbf{a} - 2\mathbf{b} - 5\mathbf{c})$.

Question 38. It is given that the vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular. Prove that each of the following can be written in the form $p|\mathbf{a}||\mathbf{b}|$, where p is a constant to be found in each case.

a) $|(\mathbf{a} + \mathbf{b}) \times (2\mathbf{a} - \mathbf{b})|$
b) $|(3\mathbf{a} - 2\mathbf{b}) \times (4\mathbf{a} + \mathbf{b})|$
c) $|(6\mathbf{b} - \mathbf{a}) \times (9\mathbf{a} - \mathbf{b})|$

Question 39. Show that if the vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel, then $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

Question 40. Given that $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors and $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = \mathbf{0}$, determine the possible angles between the vectors $\mathbf{a}$ and $\mathbf{b}$.

Question 41. It is given that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are vectors such that $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}$. Show that $\mathbf{a} + \mathbf{c} = k\mathbf{b}$, where k is a scalar quantity.

Question 42. It is given that the vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $|\mathbf{a} \times \mathbf{b}| = 7\sqrt{2}$, $|\mathbf{a}| = 4$ and $|\mathbf{b}| = 3\sqrt{2}$. Find the exact value of $\mathbf{a} \cdot \mathbf{b}$.

Question 43. It is given that the vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $\mathbf{a} \cdot \mathbf{b} = 10\sqrt{66}$, $|\mathbf{a}| = 25$ and $|\mathbf{b}| = 2\sqrt{3}$. Find the exact value of $|\mathbf{a} \times \mathbf{b}|$.

Question 44. It is given that the non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $|\mathbf{a}| = a$ and $|\mathbf{b}| = b$. It is also given that $|\mathbf{a} \times \mathbf{b}| = p$ and $\mathbf{a} \cdot \mathbf{b} = q$. Show that $a^2b^2 = p^2 + q^2$.

Question 45. In each case, find the value of $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$:

a) $\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

b) $\mathbf{a} = \begin{pmatrix} 4 \\ 16 \\ 12 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 3 \\ -3 \\ 1 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 4 \\ 1 \\ 4 \end{pmatrix}$

Question 46. Determine what you can tell about the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.

Question 47. It is given that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. Show that $\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$.

Question 48. It is given that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. Prove that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.

Question 49. It is given that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$. Prove that $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$.